Tentative Schedule (1)

Week	Date (Mon)	Lecture (Wed)	Lecture (Fri)	Tutorial
1	10/8	Electrophilic addition of alkenes	Electrophilic addition of alkenes	
2	17/8	Electrophilic aromatic substitution	Electrophilic aromatic substitution	
3	24/8	Oxidation/reduction	Oxidation/reduction	Tutorial 1 (Electrophilic addition of alkenes; Electrophilic aromatic substitution)
4	31/8	Alkene Synthesis	Enolate Chemistry (alkylation reaction)	Tutorial 2 (Electrophilic aromatic substitution; Oxidation/reduction)
5	7/9	Enolate Chemistry (Aldol reactions)	Enolate Chemistry (Aldol reactions; Claisen)	Tutorial3 (Alkene Synthesis, Enolate Chemistry-alkylation reaction)
6	14/9	Enolate Chemistry (Enol and enamine)	**Test 1** (covers weeks 1 to 5)	

Tentative Schedule (2)

Week	Date (Mon)	Lecture (Wed)	Lecture (Fri)	Tutorial
7	28/9	Conjugate addition	Review Test 1; Protecting groups	Tutorial 4 (Enol and enamine, conjugated addition) **Fri**
8	5/10	Cycloadditions	NUS wellbeing day	Tutorial 4 (Enol and enamine, conjugated addition) **Tue**
9	12/10	Cycloadditions	Sigmatropic rearrangements	Tutorial 5 (cycloaddition) **Fri**
10	19/10	Electrocyclic reaction	Radical chemistry	Tutorial 5 (cycloaddition) **Tue** Tutorial 6 (sigmatropic; electrocyclic) **Fri**
11	26/10	Radical chemistry	**Test 2** (covers week 6 to week 10, excluding radical chemistry)	Tutorial 6 (sigmatropic; electrocyclic) **Tue**
12	2/11	Radical chemistry/Cross-coupling reactions	Cross-coupling reactions	
13	9/11 Deepavali	Review Test 2; Practice (Radicals, Cross-coupling etc.)	(consultation)	

CM3121 – Advanced Organic Chemistry

Dr. Zhu Ye

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Electrophilic Addition to Alkenes (part 1)

The aims of this chapter are to …..

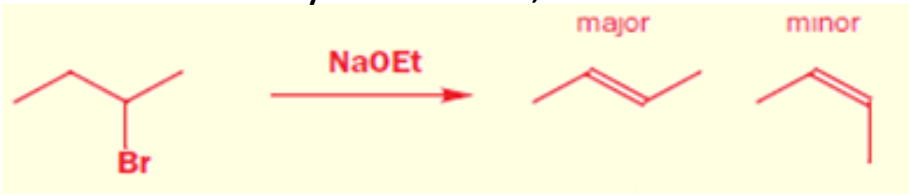
- Converting C=C double bonds to other functional groups by electrophilic addition

- How to predict which end of an unsymmetrical alkene reacts with the electrophile (regioselectivity)

- How to make alkyl halides, epoxides, alcohols, and ethers through electrophilic addition

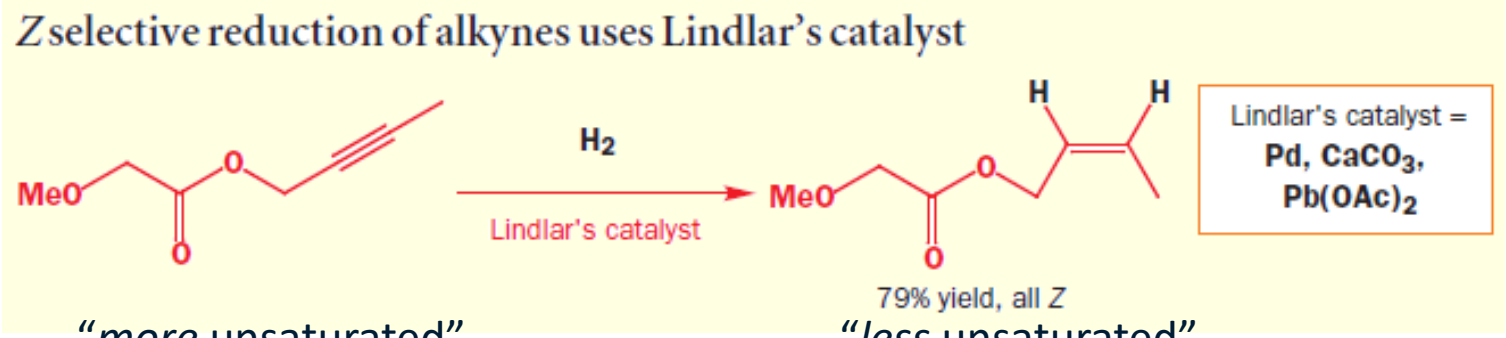
Elimination reaction and addition reaction (Synthesis of E- and Z-alkenes as an example)

- Elimination reaction of alkyl bromides, E-alkenes as the major products



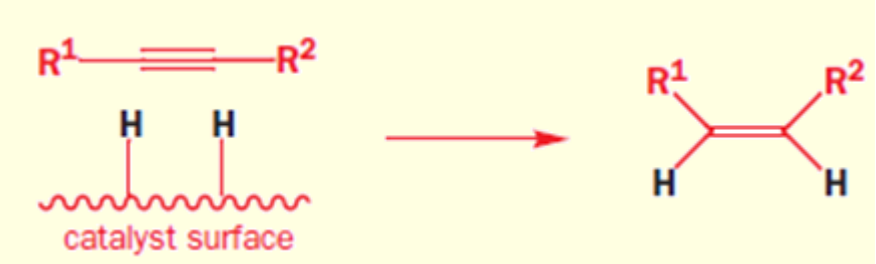
"saturated" "unsaturated"

- Addition reaction of hydrogen to alkynes



"*more* unsaturated" "*less* unsaturated"

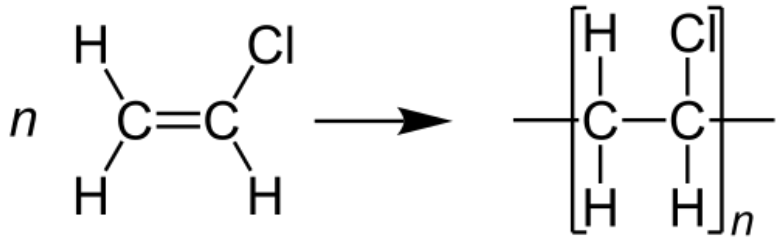
The two hydrogen atoms add to the alkyne in a *syn fashion and the alkene produced* is a *Z-alkene*

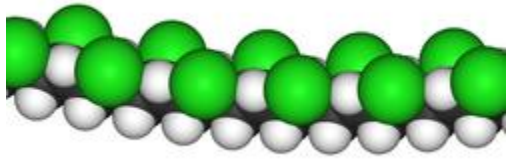


PVC (polyvinyl chloride)



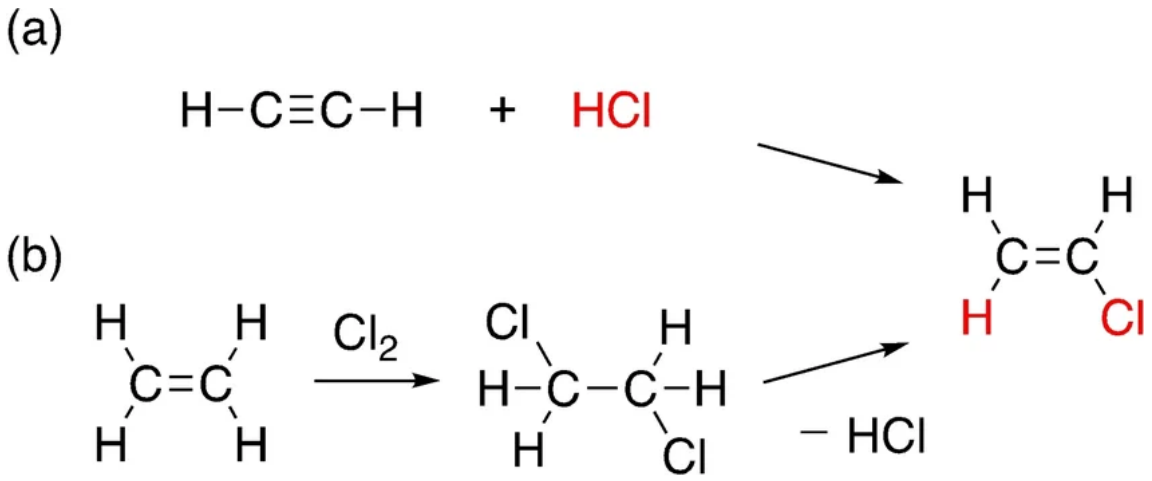
Radical chain polymerization



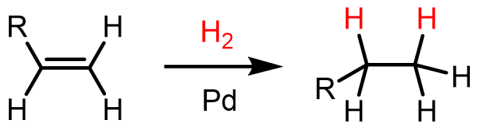




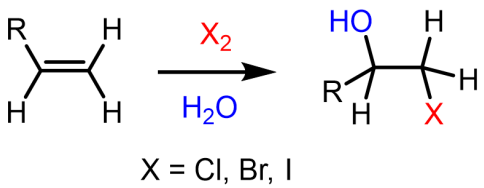
Vinyl chloride synthesis



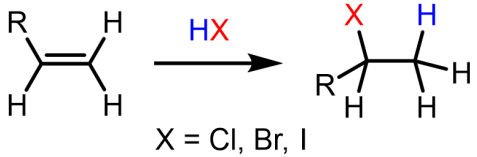
Electrophilic Addition Reactions



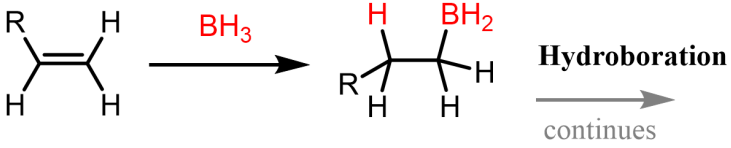
Hydrogenation

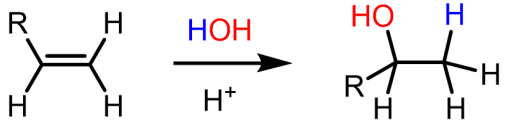


Halohydrin formation

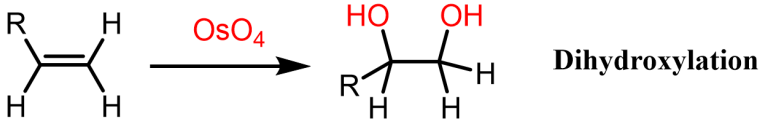


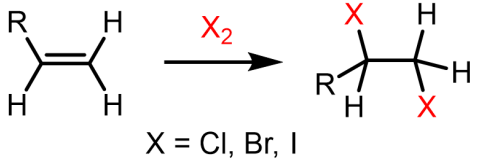
Hydrohalogenation



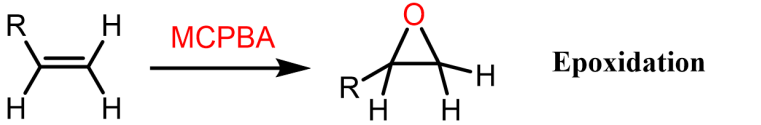


Hydration



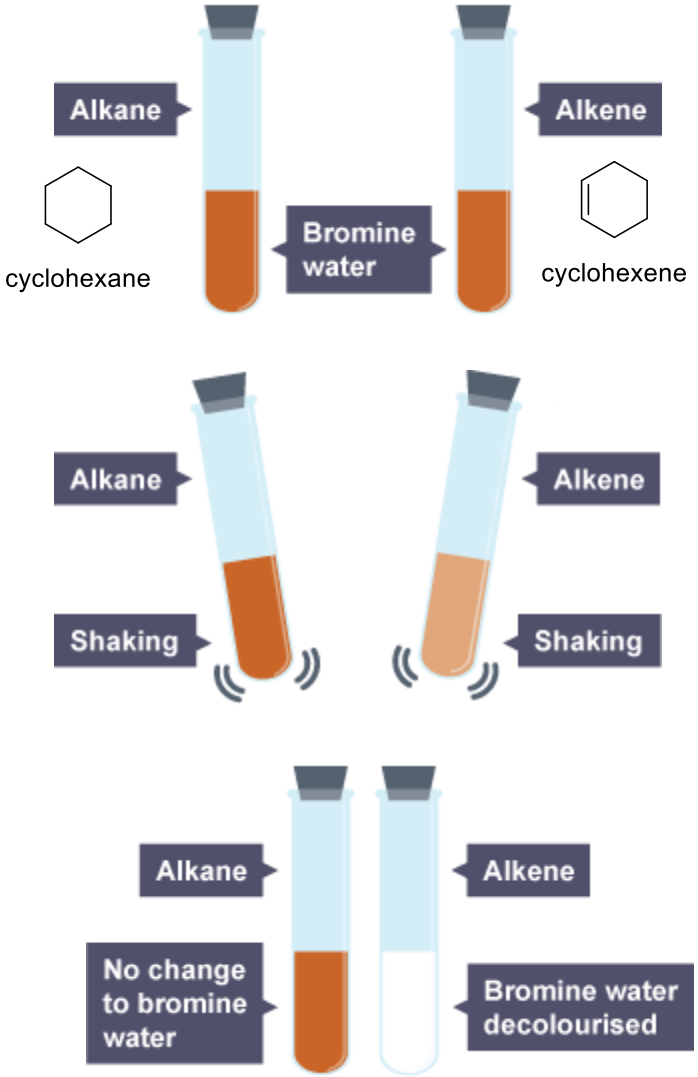


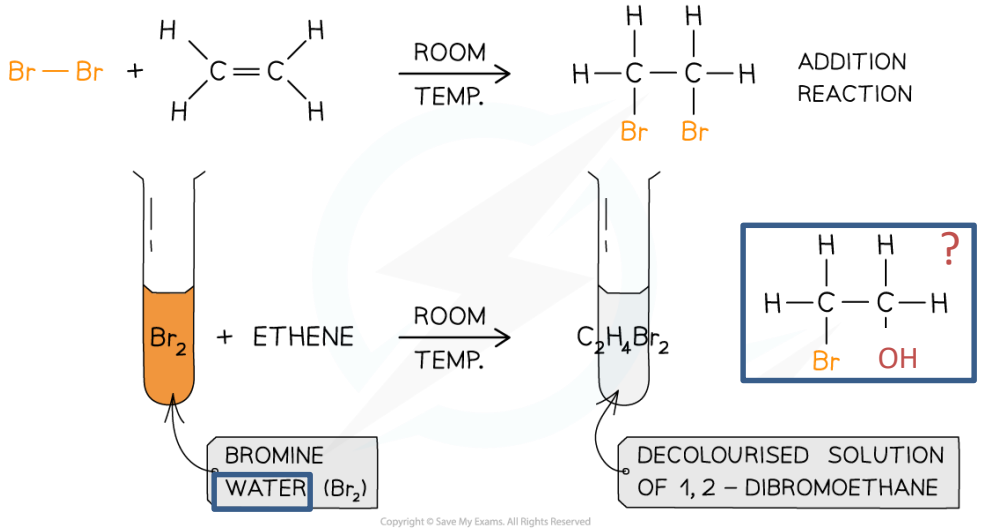
Halogenation

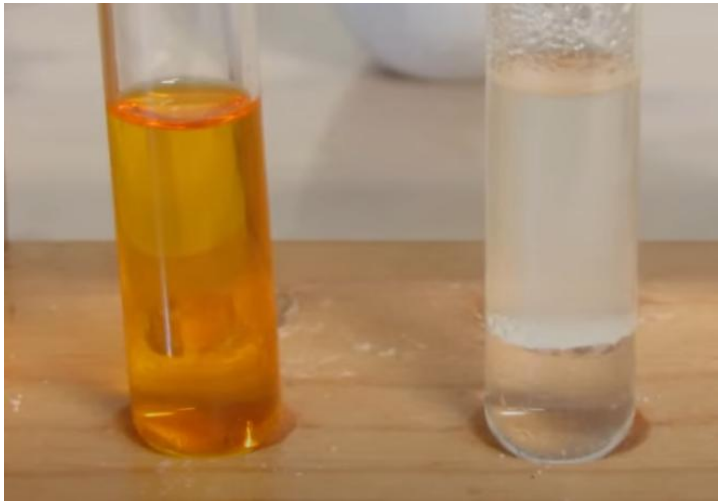


1. Alkenes react with bromine

Testing for alkenes

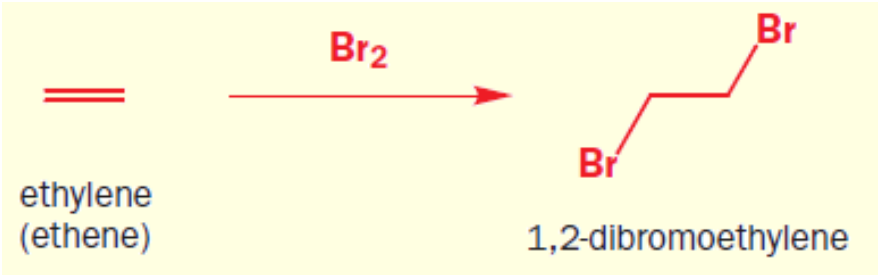






https://www.bbc.co.uk/bitesize/guides/zb6bcj6/revision/3

Reaction mechanism of halogenation of alkenes



Br–Br bond is weak, and bromine reacts with nucleophiles like this:

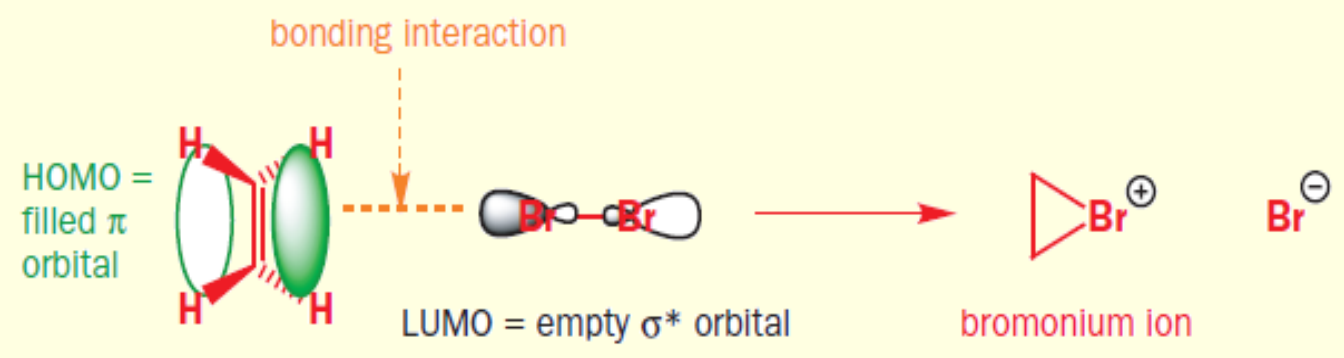


alkene must be the nucleophile, and its HOMO is the C=C π bond

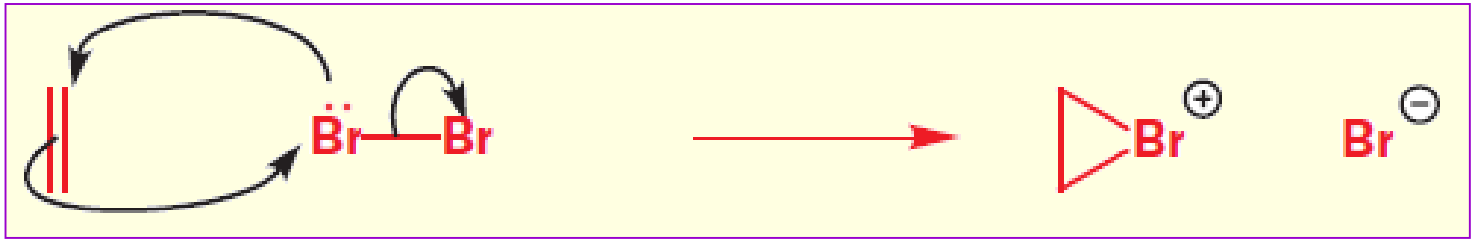


The bromonium ion as a reaction intermediate

Step 1: formation of bromonium ion

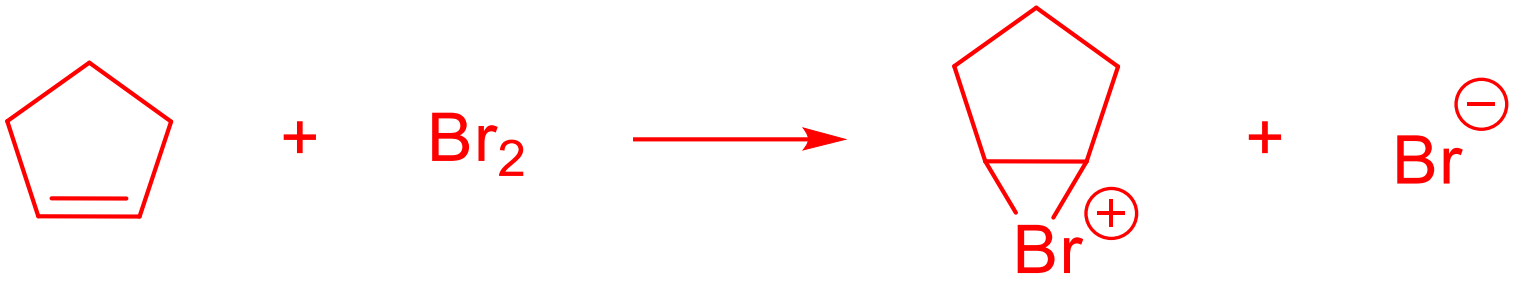


The highest electron density in the π orbital is right in the middle, between the two carbon atoms, so this is where we expect the bromine to attack. The only way the π HOMO can interact in a bonding manner with the σ* LUMO is if the Br_2 approaches end-on.

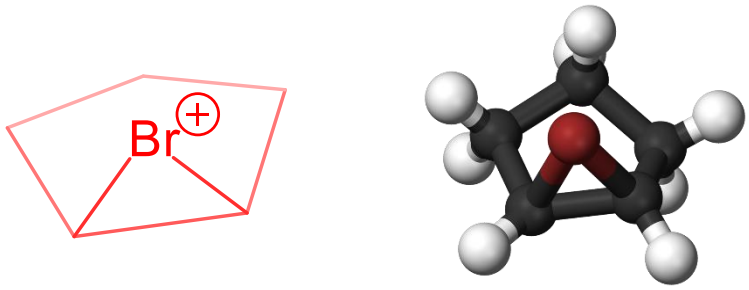


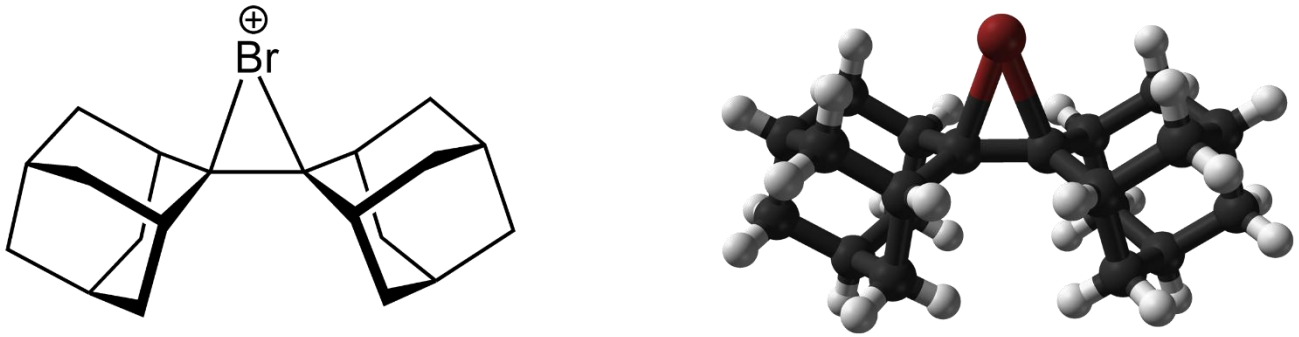
electrophilic addition to the double bond, and bromine is an electrophile.

Bromonium cations



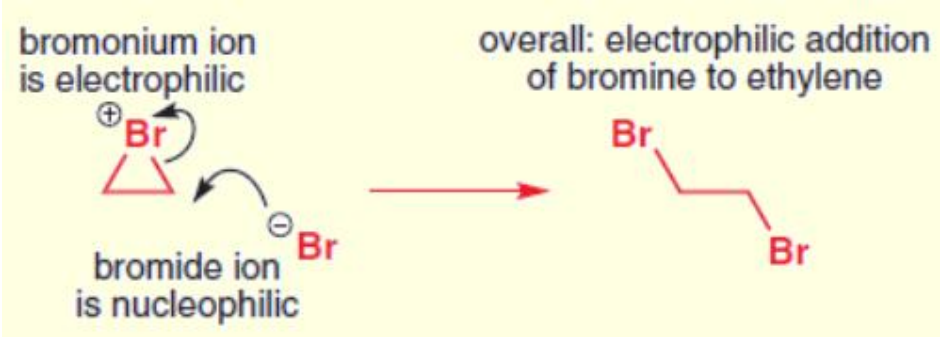
bromonium ion formed
from cyclopentene





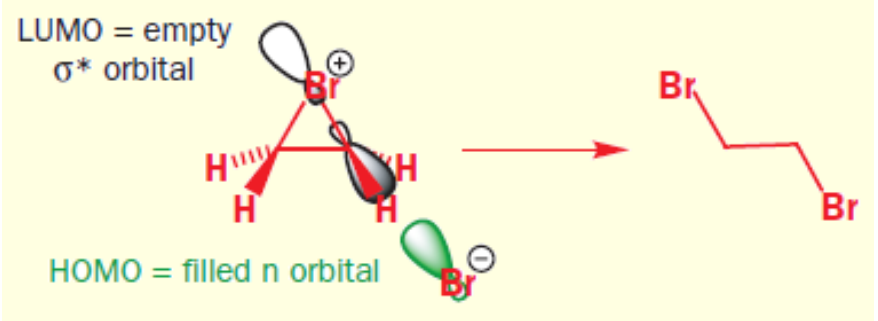
bi(adamantylidene)-derived bromonium cation (stable)

Step 2: reaction between bromonium ion and bromide



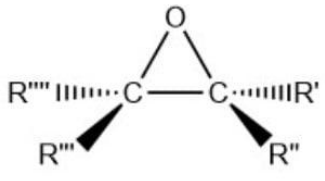
bromonium ion is an electrophile, and it reacts with the bromide ion lost from the bromine in the addition step

Attack of Br^- on a bromonium ion is a normal S_N2 substitution, hence the nucleophile maintains maximal overlap with the σ* of one of the two C–Br bonds by approaching in line with the leaving group but from the opposite side, resulting in inversion at the carbon that is attacked.



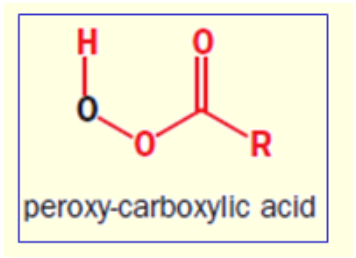
2. Oxidation of Alkenes to Form Epoxides

Drawing similarity: We can view epoxides as the oxygen analogues of bromonium ions.
Unlike bromonium ions, epoxides are quite stable.

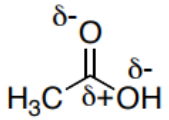


Most commonly used epoxidizing agents are peroxy-carboxylic acids. **Peroxy-acids (or peracids) have an extra oxygen atom between the carbonyl group and** their acidic hydrogen

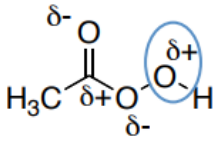
Peracids are electrophilic at oxygen, because attack there by a nucleophile displaces carboxylate, a good leaving group.



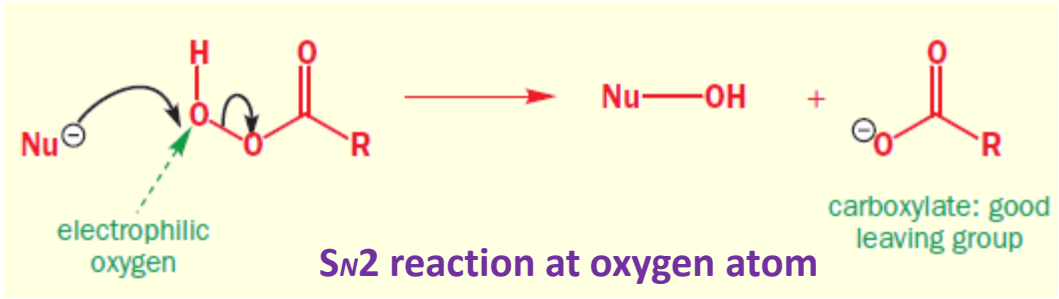
peroxy-carboxylic acid



Acetic acid



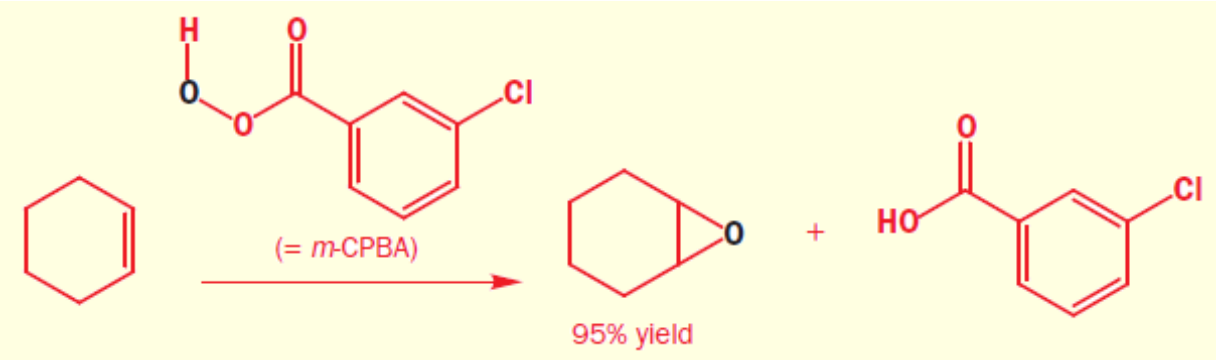
Peracetic acid
(called peracid or peroxy acid)



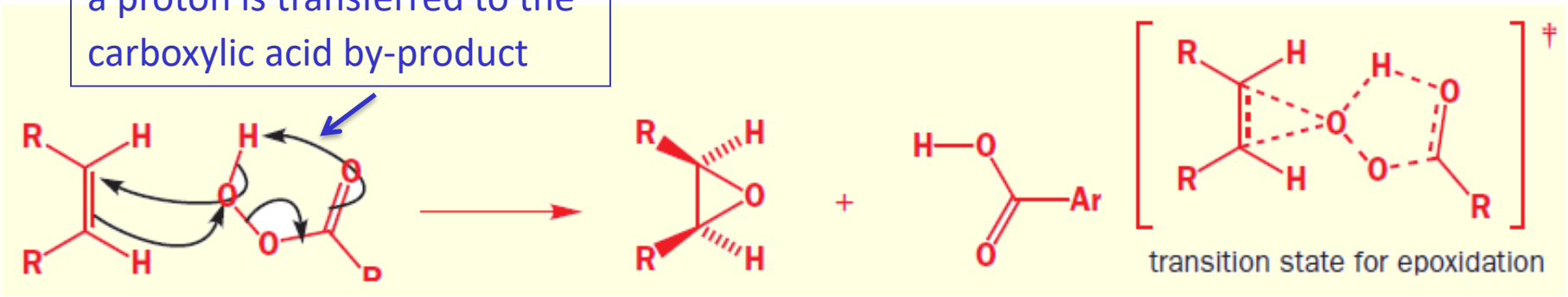
SN2 reaction at oxygen atom

m-CPBA as the oxidant

The most commonly used peroxy-acid is known as *meta*-ChloroPeroxyBenzoic Acid or *m-CPBA* because it is a safely crystalline solid.



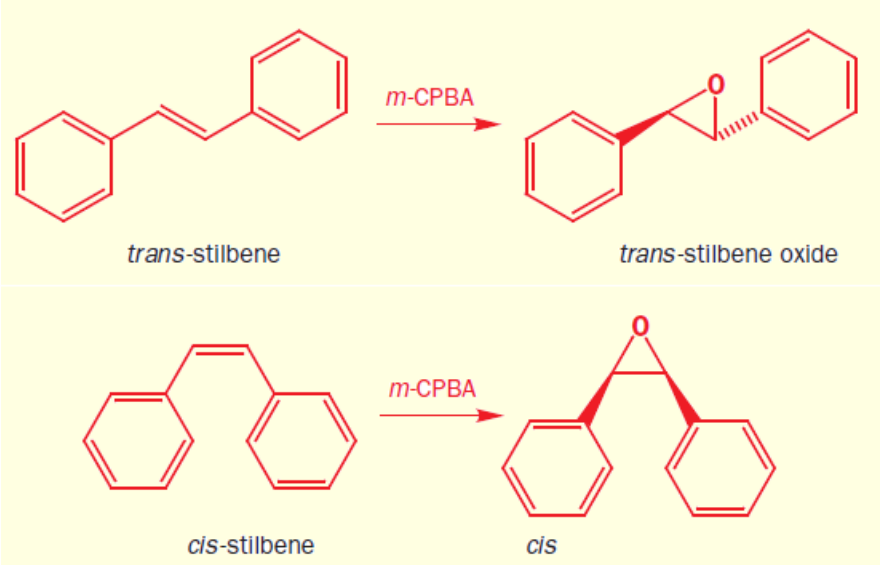
a proton is transferred to the carboxylic acid by-product



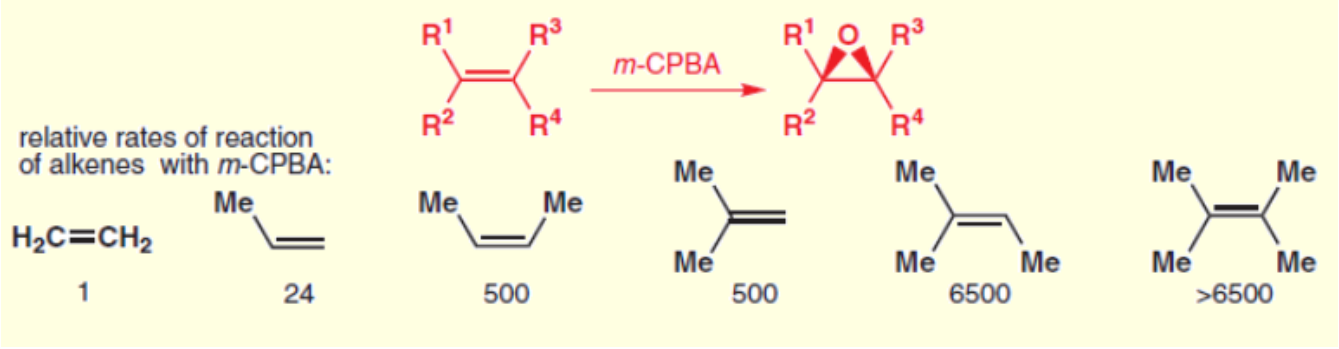
nucleophilic π bond contributes electrons to the oxygen of the weak, electrophilic polarized O–O bond

Epoxidation is Stereospecific And More Substituted Alkenes Epoxidize Faster

Since both new C–O bonds are formed on the **same face** of the alkene's π bond, the geometry of the alkene is reflected in the stereochemistry of the epoxide. The reaction is therefore stereospecific



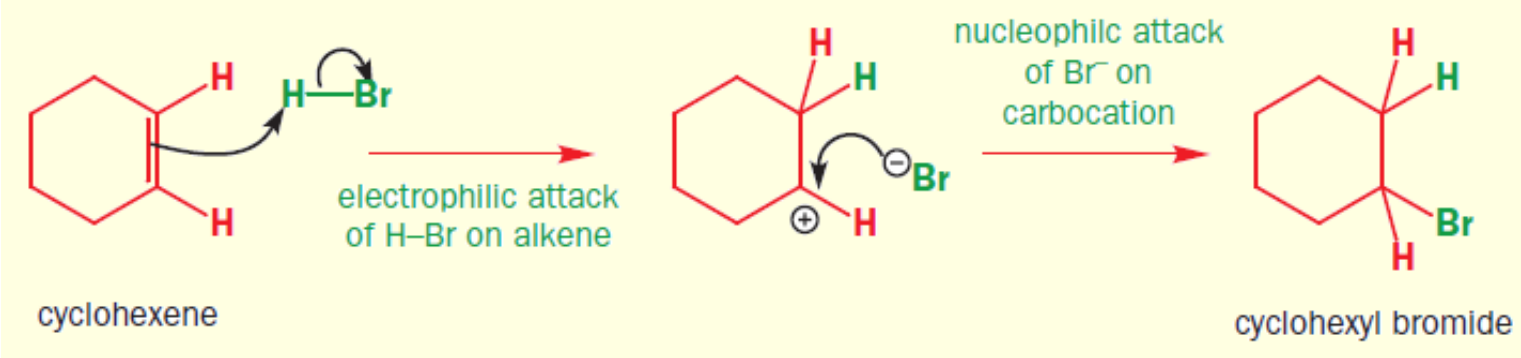
(more on stereochemistry in the next lecture)

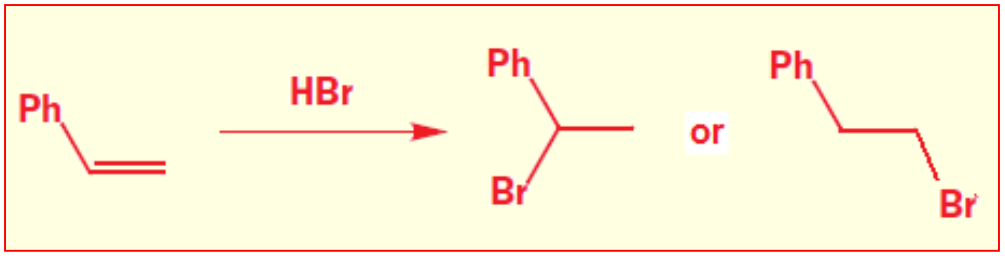


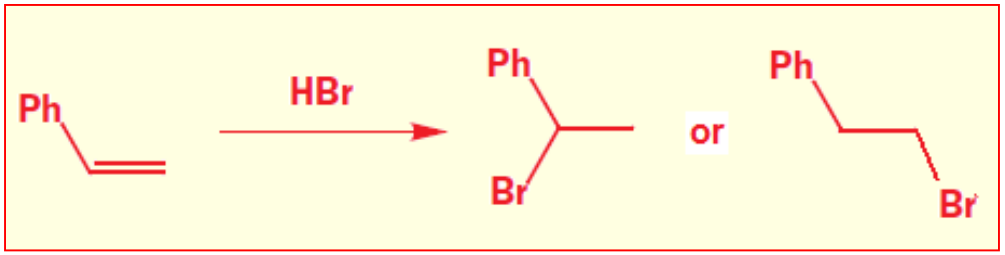
More substituted double bonds are also more nucleophilic because alkyl groups are electron-donating.

3. Electrophilic Addition of HBr to Alkenes

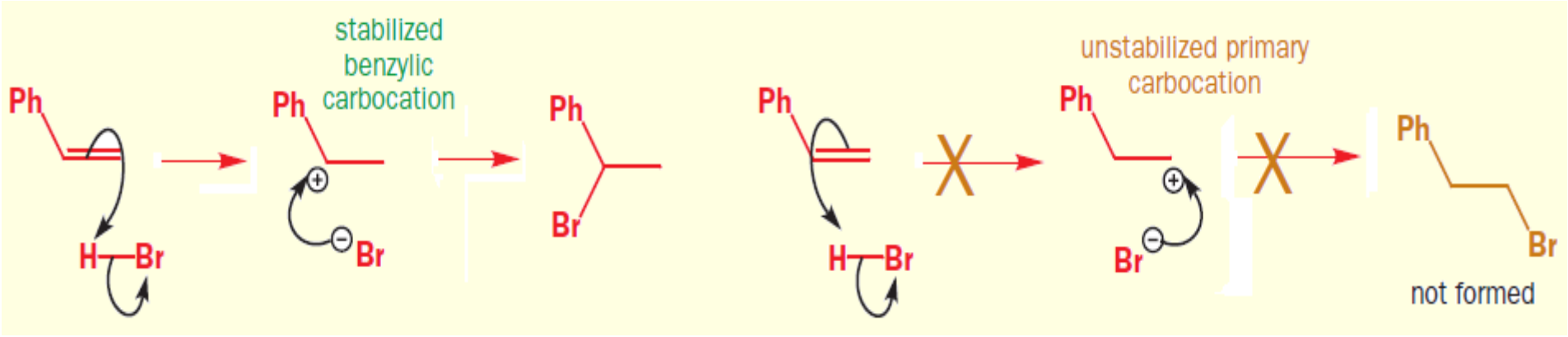
Hydrohalogenation reactions are formally the reverse of elimination reactions



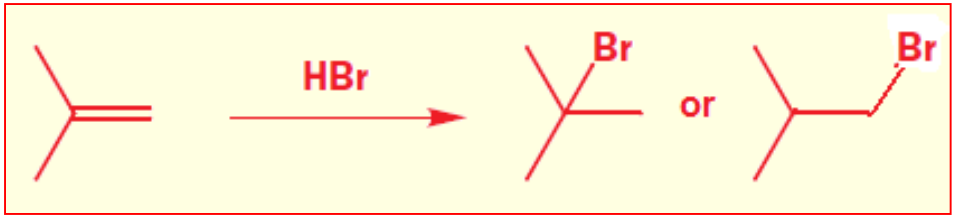


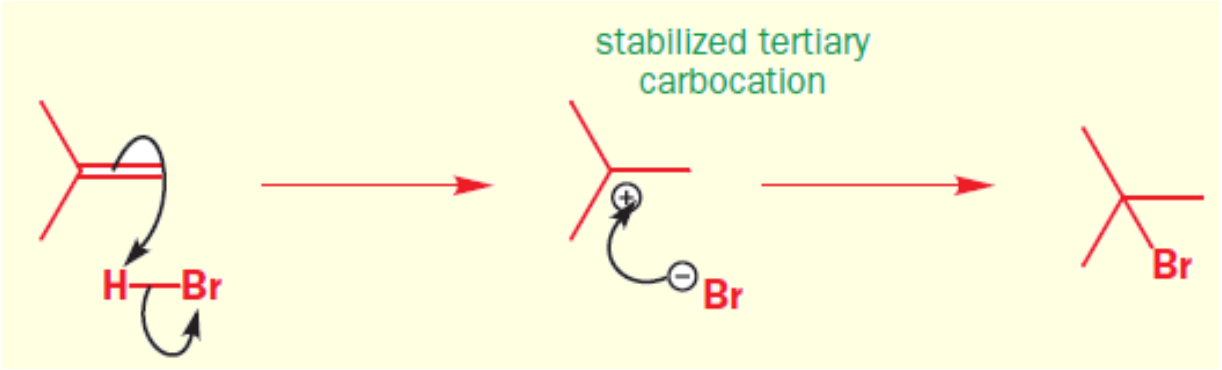


Regioselectivity: bromine atom ends up on the **more substituted** carbon



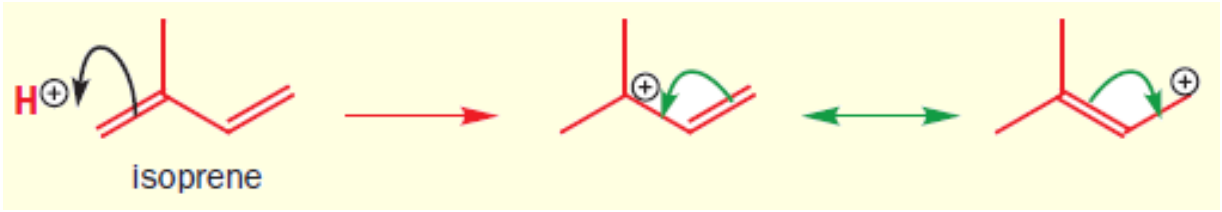
Carbocation as the reaction intermediate and Markovnikov's rule





Markovnikov's rule: The hydrogen ends up attached to the carbon of the double bond that had more hydrogens to start with.

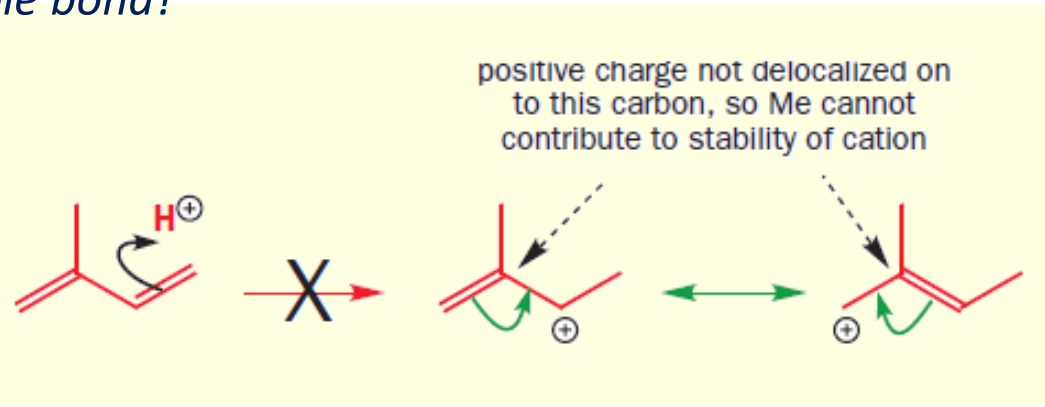
4. Electrophilic Addition to Dienes (1,4-addition)

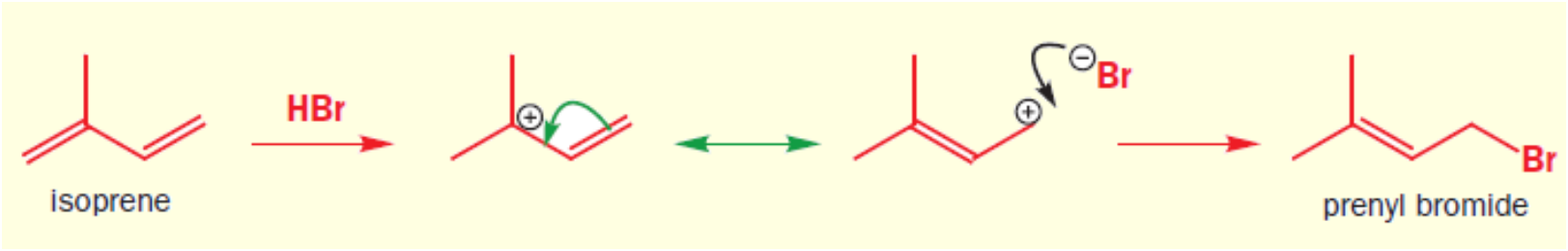


protonation gives a stable delocalized allylic cation

Why not protonate the other double bond?

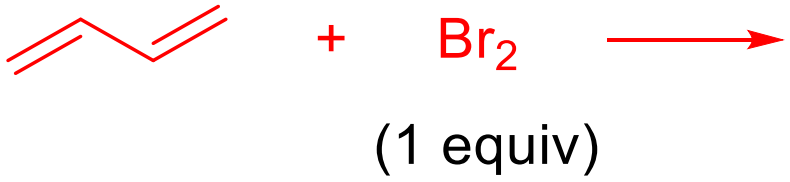
Cation obtained by protonating the other double bond is also allylic, but it cannot benefit from the additional stabilization from the methyl group





Overall, the atoms H and Br are added to the ends of the diene system

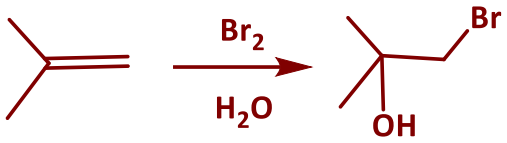
cation is attacked at the less hindered end


+
Br_2
(1 equiv)

5. Regioselectivity involving Unsymmetrical Bromonium Ions

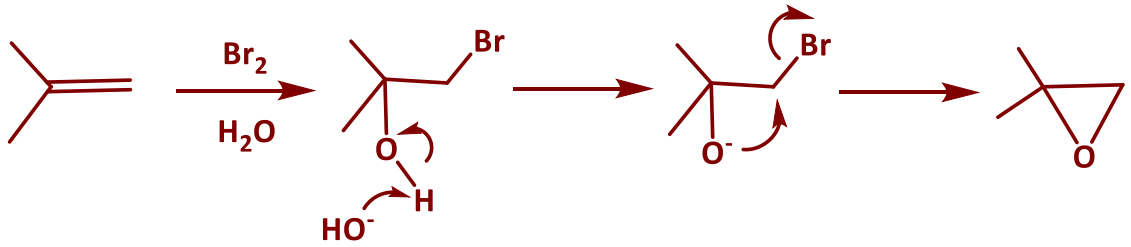
When a bromination is done in a nucleophilic solvent - eg water or methanol, solvent molecules compete with the bromide to open the bromonium ion.

Because the concentration of solvent is so high, the solvent molecule reacts first most of the time.



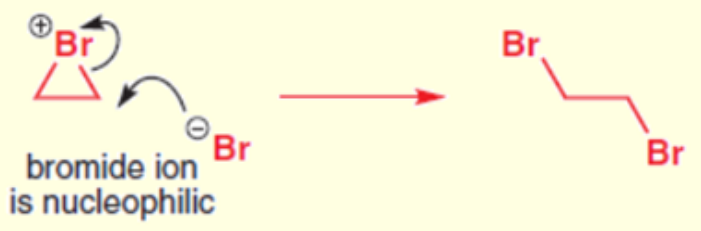
bromohydrin

The bromohydrin can serve as a precursor to the epoxide.

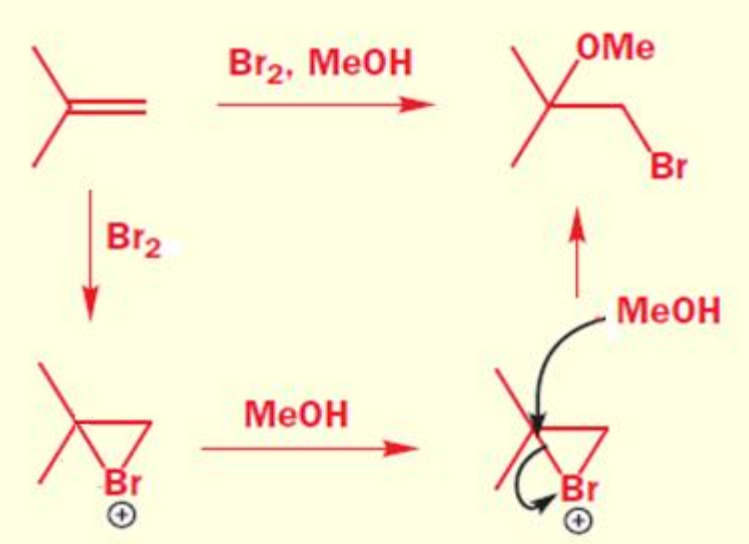


Regioselectivity

The regioselectivity is not important in the case of symmetrical bromonium ions.

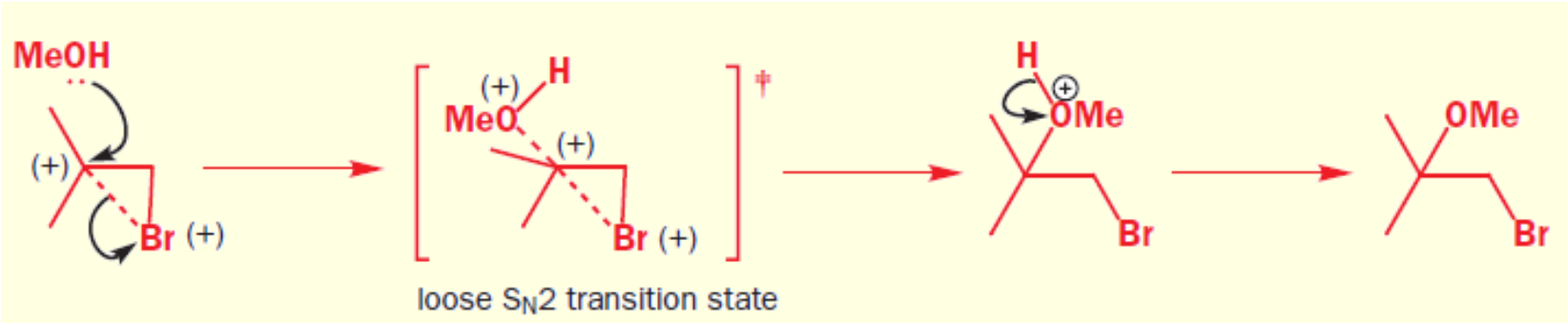


What about reactions involving **unsymmetrical Bromonium Ions?**

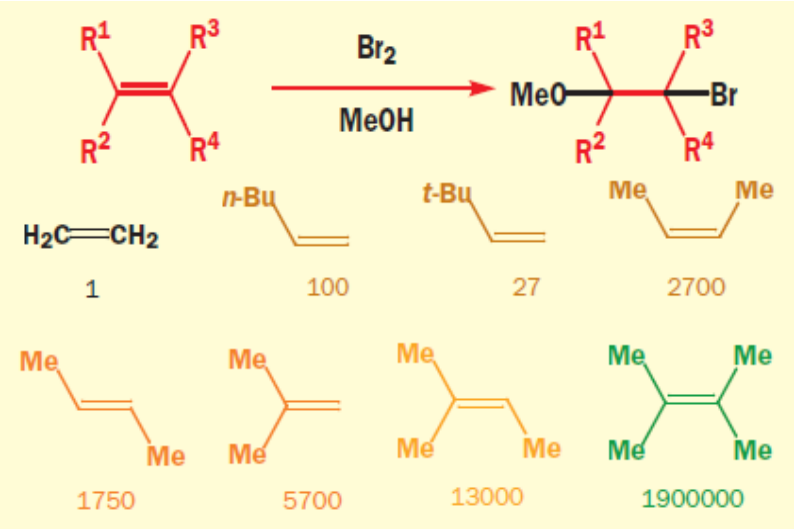


Observation:
An ether is formed by attack of methanol at the *more substituted* end of the bromonium ion.

In this case, the regioselectivity is determined by the electronic effects



bromine begins to leave, and a partial positive charge builds up at carbon. The departure of bromine can get to a more advanced state at the tertiary end than at the primary end, because the substituents stabilize the build-up of positive charge.



Rate of reactions using substituted alkenes

Summary

- Electrophilic addition reactions of alkenes are a class of reactions the convert C=C double bonds to functionalized molecules.
- Halogenation
- Epoxidation
- Hydrohalogenation
- Halohydrin (and haloether) formation
- Reaction mechanisms and intermediates
- Reaction rates and regioselectivity using substituted alkenes